A matrix-valued von Mangoldt measure in the finite Connes–van Suijlekom path

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Abstract

Fix a Galerkin level N in the finite Connes–van Suijlekom truncation of the Weil quadratic form (with no archimedean cutoff), and let $u = \log c$ be the prime cutoff, which is retained and varied. The object of this paper is the singular part of $D_u^2 Q_N$ along the finite matrix path $u \mapsto Q_N(u)$: a negative semidefinite rank-one *matrix-valued von Mangoldt measure*, a finite, cutoff-free realization of the prime side of the Weil–Guinand explicit formula on the operator path itself. Its defining computation is the first-derivative jump at a prime-power threshold $u = \log q$, which returns the von Mangoldt weight exactly, in every matrix entry:

$$Q_N'(\log q + 0) - Q_N'(\log q - 0) = -\frac{2\Lambda(q)}{\sqrt{q}\,\log q}\,\mathbf{1}_N\mathbf{1}_N^t.$$

This identity is an elementary structural derivative of the path's defining prime sum: the jump is one differentiation of the entering prime atom, and that is precisely what makes the readout exact rather than asymptotic; the identity relating the prime side to the zeros is classical and is not proven here. The contribution is the isolation and naming of the finite matrix-valued measure and the exact finite geometry established around it. First, the event is *arithmetically rigid*: the entire singular jet at an edge is a universal matrix scaled by the single scalar $\Lambda(q)$, so it carries exactly the von Mangoldt weight and nothing more. Second, the event geometry is a sharp piece of finite harmonic analysis: the source-to-jet map is a lossless finite dictionary on the squared-integer nodes $\{0, 1^2, \ldots, N^2\}$, with an explicit confluent-Vandermonde determinant, a sharp $2N{+}1$ window, and a universal recurrence $S\prod_{k=1}^{N}(S-k^2)^2$; a sharp finite vanishing-moment bound (a *prime-edge uncertainty principle* in the band-limited sense) bounds the order at which a band-limited normal can be blind to an entering prime, with the centered finite-difference stencil the unique extremizer; and each prime event is a rank-one negative-semidefinite deflation of the spectral velocity, interlacing the velocity spectrum. The statements are finite-dimensional and proved under explicitly stated hypotheses; the event algebra transfers to any L-function of the Selberg class once a finite path with the same event structure is prescribed (the arithmetic entering only through the source weights), while the sign-definite consequences require a positive weight and so hold for ζ (the spectral-barrier statement remaining conditional on its positive-definiteness hypothesis). The results are structural; the paper proves no positivity, no Riemann Hypothesis, no prime-counting, next-prime, or factoring statement.

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Note on this version. This version corrects version 1 of the Zenodo deposit: the coincidence-averaging corollary now carries its uncorrelatedness hypothesis explicitly (with the covariance discussion in Remark 3.5); the directional-residual and singular-ratio diagnostics are no longer called equivalent;

the coincidence-resolvent pole statement is stated by spectral projection, allowing removable points; the reciprocal Weyl identity is stated meromorphically with its pointwise domain made exact; and the Krein-string boundary-mass reading is downgraded from a theorem to an explicitly labeled analogy. Scope language is also recalibrated relative to version 1: the abstract's exactness sentence is removed, the Selberg-class transfer is conditioned on an actually constructed finite path, the spectral-barrier statement is demoted to a conditional remark stating its positive-definiteness hypothesis, one proof argument in the resolution-scale corollary is repaired (a domination argument replaces a first-positive-root phrase), the verification table marks the variance reduction as an IID-model statement, the vanishing-moment result is presented under a calibrated name (with the uncertainty-principle reading kept as a gloss), the related-work discussion is extended (Makraini, Andrade, Andrews), and point-of-use attributions are added. The central rank-one prime-power jump, the distributional matrix-valued measure, the dictionary, vanishing-moment, and deflation results are unchanged in statement.

1 Introduction

The Connes–van Suijlekom and Connes–Consani–Moscovici constructions attach finite Galerkin matrices to the truncated Weil quadratic form [10, 9]. Paper 1 of this sequence implemented the finite Connes–van Suijlekom computation and zero recovery numerically [16]; Paper 2 isolates the cutoff-free finite dictionary from Galerkin coefficient vectors to compactly supported Guinand–Weil test functions and proves the finite archimedean tail-order rule [15]. That cutoff-free finite matrix is the object studied here. We fix N and vary the prime cutoff $c = e^u$.

The most concrete fact is the following. At a prime-power threshold $u = \log q$, $q = p^a$, a new prime source enters the finite prime sum. It is invisible in the matrix value itself, but it is exactly visible in the first derivative, and *any one matrix entry recovers the von Mangoldt weight*,

$$\Lambda(q) = -\frac{\sqrt{q}\,\log q}{2}\left(Q_N'(\log q + 0) - Q_N'(\log q - 0)\right)_{mn} \qquad \text{for every } m, n.$$

This identity is elementary and holds by construction: the prime sum is a defining summand of Q_N, and the jump is one chain-rule differentiation of the entering atom (Section 3). The exactness is the point, and the contribution of the paper is what is isolated around it: the finite matrix-valued measure itself, its rigidity, the exact squared-node dictionary, and the sharp vanishing-moment ceiling with its unique extremizer. Summing these jumps against a test function reproduces $\sum_q \Lambda(q)\phi(\log q)$, the prime side of the Weil–Guinand explicit formula [26, 17]; the archimedean and pole terms (Paper 2) carry the complementary side, and the eigenvalues of Q_N approximate the zeros (Paper 1; a quantitative convergence law for the first zero in the CCM setting is given by Andrews [5]). In this sense the finite path renders the *prime side* of the explicit formula exactly and without an archimedean cutoff, with the primes appearing as the singular part of a second derivative; the archimedean and pole side is closed-form (Paper 2) and the zeros enter only approximately (Paper 1). Morán Ledezma gives a probabilistic "arithmetic spectral measure" reading of the explicit sums [23]; the object here is the finite matrix-valued measure carried by the Connes–van Suijlekom path, distinct from that scalar construction.

The rest of the arithmetic story is a rigidity theorem: beyond the scalar $\Lambda(q)$, the local event contains no further information about q, and the universal singular jet has an explicit closed form (Section 4). The redundancy of the rank-one measure has a concrete payoff: under mean-zero, equal-variance, pairwise uncorrelated entrywise errors, averaging its $(2N+1)^2$ entries reads $\Lambda(q)$ with variance reduced by the same factor (Corollary 3.4; the covariance hypothesis is essential, Remark 3.5). What remains is the finite algebra of the event geometry, recorded as a reference catalog. The finite source-to-jet map is a lossless dictionary whose nodes are the squares $0, 1^2, \ldots, N^2$ (Section 5); a sharp vanishing-moment ceiling governs how blind a band-limited probe can be to an entering prime, with a unique extremizer (Section 6); and the rank-one event deflates the spectral velocity, interlaces its spectrum, and is generated by a single coincidence resolvent (Section 7). The event algebra transfers across the Selberg class conditionally on the corresponding finite paths, with a residue-class readout of $\Lambda(q)$ across the Dirichlet family under the same hypothesis; the sign-definite parts require a positive weight (Section 8).

Here an edge is treated only after a nonsmooth point of the path has been localized; nothing below predicts future prime powers from sampled data, and nothing below is a positivity statement. To our knowledge the finite Connes–van Suijlekom matrix-valued von Mangoldt measure and its rigidity, together with the squared-node event dictionary and the finite prime-edge vanishing-moment ceiling, have not previously been isolated. The trace-formula setting is due to Connes [7, 8]; the finite Galerkin framework to Connes–van Suijlekom and Connes–Consani–Moscovici; Suzuki develops a continuous screw-function framework [25]; Śliwiński studies the adjacent $D_{\log}^{(\lambda,N)}$ spectral-error regime [24]; Makraini records unconditional spectral bounds, a Krein-space symmetry, and a norm-convergence obstruction for the truncated Weil operator compressed to a prolate Sonin space, reducing positivity in that framework to an asymptotic kernel-invisibility property [22], a line concerning positivity and convergence in the cutoff rather than the derivative structure of a fixed-N path; Andrade records related Loewner, exact-entry, operator-monotone, and rank-one pole-resolvent framings [2, 3, 1, 4]; and Liao uses prime-event language for statistical event sequences [20]. The classical Prony and confluent-Vandermonde algebra [12, 13, 6], the explicit formula [26, 17], and the vanishing-moment (sum-rule) inequality [11] are used as tools; what is new, to our knowledge, is their exact realization on the finite Connes–van Suijlekom path.

2 The finite path

Fix $N \geq 0$ and write

$$I_N = \{-N, \ldots, N\}, \qquad d = 2N+1, \qquad \mathbf{1}_N = (1)_{m \in I_N}.$$

For a single prime source at reduced position $\omega \in [0,1]$, the finite Galerkin matrix of that source is the divided-difference matrix $A_N(\omega)$ on I_N,

$$(A_N(\omega))_{mn} = \begin{cases} \dfrac{\sin(2\pi m\omega) - \sin(2\pi n\omega)}{\pi(m-n)}, & m \neq n, \\ 2\omega\cos(2\pi m\omega), & m = n, \end{cases}$$

in the normalization of the finite dictionary [15]. A symbol-level Loewner divided-difference representation of the prime contribution in the localized Weil form was recorded earlier by Andrade [2]; what is used here is the derivative jump of the path across the threshold, not the representation itself. Its off-diagonal entries are divided differences of $m \mapsto \sin(2\pi m\omega)$; this is why a source entering at $\omega = 0$ is invisible in the matrix value but exact in the first derivative (Lemma 2.3).

Remark 2.1 (Divided-difference structure). Write $\mathrm{DD}_X[g]$ for the confluent divided-difference matrix of a C^1 function g on real nodes $X = \{x_m\}$, with entries $(g(x_m) - g(x_n))/(x_m - x_n)$ off the diagonal and $g'(x_m)$ on it. Then $A_N(\omega) = \mathrm{DD}_X[g_\omega]$ with $X = \{-N, \ldots, N\}$ and $g_\omega(x) = \sin(2\pi x\omega)/\pi$: the diagonal $2\omega\cos(2\pi m\omega)$ is exactly the confluent limit $g'_\omega(m)$. This is the divided-difference representation of the finite quadratic form established in [10, Props. 4.1–4.2]; here it is specialized to a single prime source and differentiated in the prime cutoff along the path. The whole event package below uses only this structure. Indeed, for any real-analytic background $R(u)$, amplitude $s(u)$, and reduced position $\omega(u)$ with $\omega(u_0) = 0$, $\omega'(u_0) \neq 0$, such that g_0 is constant on the node span (so $\mathrm{DD}_X[g_0] = 0$) and $\frac{d}{d\omega} g_\omega|_0$ is affine on the nodes with nonzero slope λ, the path $u \mapsto R(u) - s(u)\,\mathrm{DD}_X[g_{\omega(u)}]\,\mathbf{1}[u \geq u_0]$, with the source term present only for $u \geq u_0$ (the entering-source mechanism of Theorem 3.1), has the universal rank-one first-derivative jump $-s(u_0)\omega'(u_0)\lambda\,\mathbf{1}_N\mathbf{1}_N^t$ and the ensuing rigidity, independent of the node geometry. The finite Connes–van Suijlekom path is the case $g_\omega = \sin(2\pi x\omega)/\pi$ ($g_0 = 0$, $\lambda = 2$), $s = \Lambda(q)/\sqrt{q}$, $\omega = 1 - \log q/u$.

Let $Q_N(u)$ denote the cutoff-free finite Connes–van Suijlekom/Connes–Consani–Moscovici matrix at prime cutoff $c = e^u$:

$$Q_N(u) = Q_{\mathrm{arch}}(u) + Q_{\mathrm{pole}}(u) - \sum_{q=p^a \leq e^u} \frac{\Lambda(q)}{\sqrt{q}}\, A_N\!\left(1 - \frac{\log q}{u}\right).$$

Here Q_{arch} and Q_{pole} are the closed-form finite archimedean and pole matrices of the finite dictionary, given explicitly in Paper 2 [15, Lemma 2.1 and the displayed pole and archimedean sources]: Q_{pole} is the divided-difference matrix of the pole source ψ_0, and Q_{arch} that of the archimedean density, both in the same A_N-normalization as the prime blocks. We call the path *archimedean-cutoff-free* because no archimedean cutoff is present: only the prime cutoff e^u is varied. This is the standard finite Connes–van Suijlekom/Connes–Consani–Moscovici assembly with the prime cutoff exposed as a variable, not an ad hoc signal. From Paper 2 we use only the following regularity of the non-prime blocks.

Lemma 2.2 (Background regularity, imported from Paper 2)**.** *The archimedean and pole blocks $Q_{\mathrm{arch}}(u)$ and $Q_{\mathrm{pole}}(u)$ are real analytic on $(0, \infty)$. Consequently the only points at which $u \mapsto Q_N(u)$ can fail to be real analytic are the prime-power thresholds $u = \log q$.*

Proof. This lemma is imported: the blocks are defined, entrywise and in closed form, in [15, Lemma 2.1], and the analyticity below only restates the regularity of those closed forms. Entrywise, the pole block is a rational-hyperbolic expression whose denominators $u^2 + 16\pi^2 n^2$ are strictly positive for $u > 0$, and the archimedean block is built from digamma and trigamma values at $\frac{1}{4} + \frac{i\pi n}{u}$, whose arguments avoid the poles of ψ, ψ' for $u > 0$, together with a correction series of geometric ratio e^{-2u}. Each summand extends holomorphically to a complex neighborhood of $(0, \infty)$, and on such a neighborhood the e^{-2u} ratio bound gives uniform convergence on compact subsets, so each block is real analytic there [15]. Away from the finitely many thresholds in a compact interval, each reduced position $1 - \log q/u$ is real analytic in u, so Q_N is real analytic off $\{\log q\}$. □

Lemma 2.3 (Edge derivative)**.** *The single-source matrix satisfies $A_N(0) = 0$ and $A_N'(0) = 2\,\mathbf{1}_N\mathbf{1}_N^t$.*

Proof. The diagonal gives $A_N(0)_{mm} = 0$, and for $m \neq n$ the numerator vanishes at $\omega = 0$, so $A_N(0) = 0$. Differentiating at zero gives $(2\pi m - 2\pi n)/(\pi(m - n)) = 2$ for $m \neq n$, and the diagonal derivative of $2\omega\cos(2\pi m\omega)$ at zero is 2. □

A companion structural fact, used throughout Section 5, is that every source profile has only odd Taylor content at the edge.

Lemma 2.4 (Odd content)**.** *For each m, n the entry $\omega \mapsto (A_N(\omega))_{mn}$ is an odd function of ω; its even Taylor coefficients at $\omega = 0$ vanish, and all arithmetic content of the event sits in the odd jets.*

Proof. Both $\sin(2\pi m\omega)$ and $\omega\cos(2\pi m\omega)$ are odd in ω, and the off-diagonal quotient of odd functions by the constant $\pi(m - n)$ is odd. □

3 The matrix-valued von Mangoldt measure

Theorem 3.1 (Prime-power derivative jump)**.** *The path $u \mapsto Q_N(u)$ is continuous and piecewise real analytic on $(0, \infty)$, with first-derivative jumps only at $u = \log q$, $q = p^a$. At such a point,*

$$\Delta Q_N'(\log q) := Q_N'(\log q + 0) - Q_N'(\log q - 0) = -\frac{2\Lambda(q)}{\sqrt{q}\,\log q}\,\mathbf{1}_N \mathbf{1}_N^t,$$

and consequently, for every entry (m, n),

$$\Lambda(q) = -\frac{\sqrt{q}\,\log q}{2}\big(Q_N'(\log q + 0) - Q_N'(\log q - 0)\big)_{mn}.$$

Proof. By Lemma 2.2 the path is real analytic off $\{\log q\}$, hence piecewise real analytic. At $u_0 = \log q$ the entering source has $\omega_q(u) = 1 - \log q/u$, so $\omega_q(u_0) = 0$ and $\omega_q'(u_0) = 1/\log q$. Since $A_N(0) = 0$ the path is continuous at u_0, and the derivative jump comes only from the entering atom and Lemma 2.3:

$$-\frac{\Lambda(q)}{\sqrt{q}}\,A_N'(0)\,\omega_q'(u_0) = -\frac{2\Lambda(q)}{\sqrt{q}\,\log q}\,\mathbf{1}_N \mathbf{1}_N^t.$$

All other terms are analytic at u_0 and cancel. The recovery formula follows because every entry of $\mathbf{1}_N \mathbf{1}_N^t$ equals 1. $\square$

Theorem 3.1 is an elementary structural derivative of the path's defining prime sum: the prime block is a defining summand of $Q_N(u)$, and the jump is one chain-rule differentiation of the entering atom (Lemma 2.3). This is a feature, not an accident: it is what makes the readout exact rather than asymptotic. The contribution of the paper is not the differentiation but the isolation of the resulting measure (Corollary 3.2) and the exact finite geometry around it (Sections 4–7); see the scope discussion in Section 8.

Corollary 3.2 (Matrix-valued von Mangoldt measure)**.** *In the sense of distributions on $(0, \infty)$,*

$$(D_u^2 Q_N)_{\text{sing}} = -\sum_{q=p^a} \frac{2\Lambda(q)}{\sqrt{q}\,\log q}\,\mathbf{1}_N \mathbf{1}_N^t\,\delta_{\log q}.$$

Each atom is negative semidefinite and has rank one. Writing μ_{mn} for the scalar singular measure $\big((D_u^2 Q_N)_{mn}\big)_{\text{sing}} = -\sum_q \frac{2\Lambda(q)}{\sqrt{q}\,\log q}\,\delta_{\log q}$ (the same for every (m, n)), one has, for every compactly supported ϕ,

$$\sum_{q=p^a} \Lambda(q)\phi(\log q) = -\frac{1}{2}\int_0^\infty e^{u/2} u\,\phi(u)\;d\mu_{mn}(u).$$

Proof. A continuous piecewise-analytic scalar whose first derivative jumps by J at u_0 has singular second derivative $J\delta_{u_0}$; apply this entrywise to Theorem 3.1. At $u = \log q$ the factor $-e^{u/2}u/2$ converts $2\Lambda(q)/(\sqrt{q}\log q)$ back to $\Lambda(q)$. $\square$

Corollary 3.2 is the paper's central object: the finite Connes–van Suijlekom path carries the prime side of the explicit formula [26, 17] as an exact, cutoff-free, negative semidefinite rank-one *matrix-valued von Mangoldt measure*. Figure 1 shows a scalar readout of the path and the corresponding measure of rank-one shocks.

The first derivative jumps down by a rank-one shock; the second derivative jumps *up* by the complementary rank-one shock, in the same universal direction.

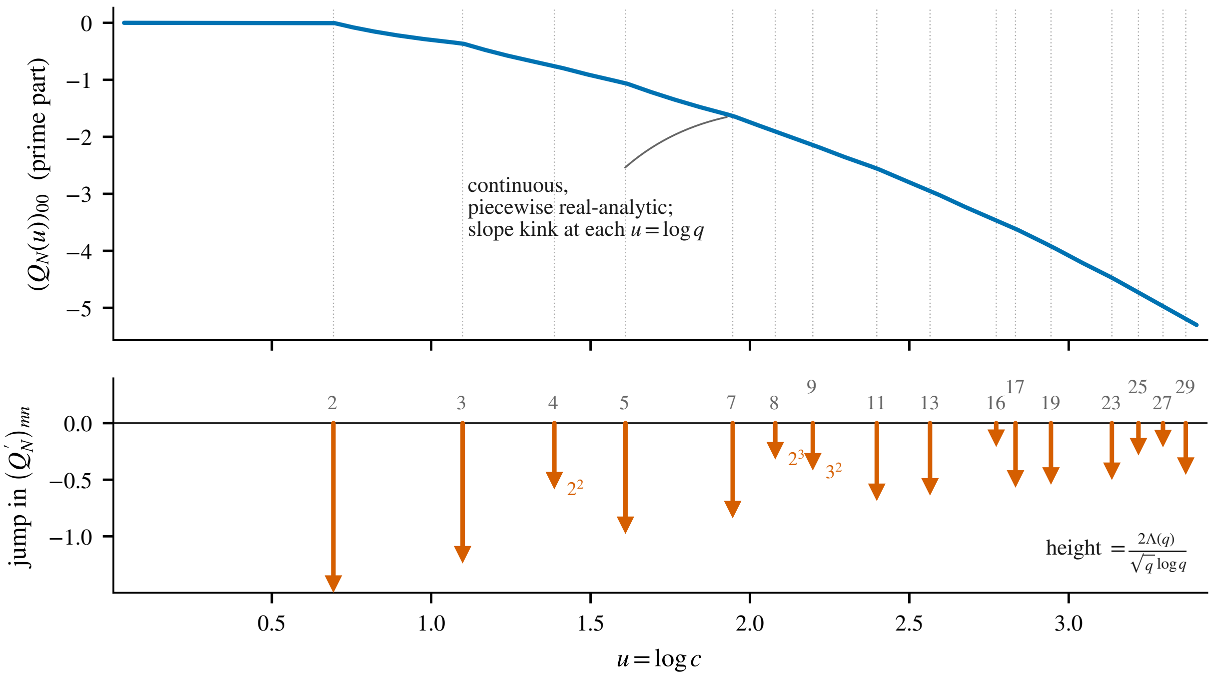

Figure 1: The finite path as an arithmetic event signal (top: a scalar entry of the prime part, continuous and piecewise real analytic, with a slope kink at each $u = \log q$) and its second-derivative singular part (bottom: the matrix-valued von Mangoldt measure, a rank-one negative shock of height $2\Lambda(q)/(\sqrt{q}\,\log q)$ at each prime power, including the true powers $4 = 2^2$, $8 = 2^3$, $9 = 3^2$).

Proposition 3.3 (Second-order event law)**.** *At a prime-power event $u_0 = \log q$ the classical (non-singular) second derivative of the path has the rank-one positive semidefinite jump*

$$Q_N''(\log q + 0) - Q_N''(\log q - 0) = +\frac{4\Lambda(q)}{\sqrt{q}\,(\log q)^2}\,\mathbf{1}_N\mathbf{1}_N^t,$$

opposite in sign to the first-derivative jump of Theorem 3.1.

Proof. By Lemma 2.2 the jump comes only from the entering atom $E_q(\tau) = -\frac{\Lambda(q)}{\sqrt{q}}A_N(\omega_q(\tau))$, $\omega_q = \tau/(u_0+\tau)$. Then $E_q''(0+) = -\frac{\Lambda(q)}{\sqrt{q}}\big(A_N''(0)\,\omega_q'(u_0)^2 + A_N'(0)\,\omega_q''(u_0)\big)$; since A_N is odd (Lemma 2.4) $A_N''(0) = 0$, and with $A_N'(0) = 2\mathbf{1}_N\mathbf{1}_N^t$, $\omega_q''(u_0) = -2/u_0^2$ this is $+\frac{4\Lambda(q)}{\sqrt{q}\,u_0^2}\mathbf{1}_N\mathbf{1}_N^t$. □

Because the jump lies entirely in the known rank-one direction $\mathbf{1}_N\mathbf{1}_N^t$, every entry of the jump matrix carries the same scalar, so the $(2N+1)^2$ entries can be pooled. Since all entries of the model jump *coincide* in the universal direction, we call the resulting all-ones compressions *coincidence* objects: the coincidence average below, and the coincidence resolvent and coincidence-Weyl function of Section 7. How much pooling helps is decided by the correlation structure of the measurement errors, which must therefore be part of the hypothesis.

Corollary 3.4 (Coincidence-averaged weight readout)**.** *Suppose the first-derivative jump is measured or estimated as $\widehat{J} = -a_q\,\mathbf{1}_N\mathbf{1}_N^t + E$, $a_q = 2\Lambda(q)/(\sqrt{q}\,\log q)$, with entrywise errors E_{mn} of mean zero and common variance σ^2 that are* pairwise uncorrelated*. Among linear unbiased estimators of*

a_q, the minimum-variance one (Gauss–Markov) is the coincidence average

$$\widehat{a}_q = -\frac{1}{(2N+1)^2}\sum_{m,n}\widehat{J}_{mn}, \qquad \mathrm{Var}(\widehat{a}_q) = \frac{\sigma^2}{(2N+1)^2},$$

a variance smaller by the factor $(2N+1)^2$ than the single-entry readout of Theorem 3.1. The residual $\widehat{J} + \widehat{a}_q \mathbf{1}_N \mathbf{1}_N^t$ is orthogonal to $\mathbf{1}_N\mathbf{1}_N^t$, and its Frobenius size measures the distance of $\widehat{J}$ to the one-dimensional model span of $\mathbf{1}_N\mathbf{1}_N^t$.

Proof. For the unit-Frobenius direction $u = \mathbf{1}_N\mathbf{1}_N^t/\|\mathbf{1}_N\mathbf{1}_N^t\|_F$, the amplitude functional $\langle u,\cdot\rangle_F$ is the Gauss–Markov estimator of the coefficient of u; with $\|\mathbf{1}_N\mathbf{1}_N^t\|_F^2 = (2N+1)^2$ this is the stated average, and for pairwise uncorrelated entries $\mathrm{Var} = \sigma^2\|\mathbf{1}_N\mathbf{1}_N^t\|_F^2/(\|\mathbf{1}_N\mathbf{1}_N^t\|_F^2)^2 = \sigma^2/(2N+1)^2$, against σ^2 for one entry. The orthogonal split $\widehat{J} = -\widehat{a}_q\mathbf{1}_N\mathbf{1}_N^t + R$, $R \perp \mathbf{1}_N\mathbf{1}_N^t$, gives the residual statement. □

Corollary 3.4 is a design upper bound on the pooling gain under an idealized error model, approximated for instance when the entrywise estimates are formed from disjoint sample sets; Remark 3.5 records the realistic correlated regimes.

Remark 3.5 (Covariance hypotheses and diagnostics). (i) The uncorrelatedness hypothesis cannot be dropped: if all entries carry one common error variable, $E_{mn} \equiv Z$ with $\mathrm{Var}(Z) = \sigma^2$, then every mean/variance hypothesis holds while $\mathrm{Var}(\widehat{a}_q) = \sigma^2$, with no reduction. For a general *nonsingular* (positive definite) error covariance Σ the minimum-variance linear unbiased readout is the generalized least-squares estimator with weights Σ^{-1}, and its variance is governed by Σ, not by the entry count; for singular Σ, as in the common-error example just given, the best linear unbiased estimator exists but is not given by this formula. (ii) If the measurement is symmetric, $E_{mn} = E_{nm}$, the entries cannot all be uncorrelated; under the natural model (the entries with $m \le n$, diagonal included, pairwise uncorrelated, each of variance σ^2) the coincidence average has variance $\sigma^2(2d-1)/d^3$ with $d = 2N+1$, a reduction of order $d^2/2$ rather than d^2: indeed $\mathrm{Var}\big(\sum_{m,n} E_{mn}\big) = d\,\sigma^2 + 4\binom{d}{2}\sigma^2 = d(2d-1)\sigma^2$, since each of the $\binom{d}{2}$ off-diagonal values appears twice in the sum, and dividing by d^4 gives the formula. (iii) The directional residual and the singular-value ratio σ_2/σ_1 of $\widehat{J}$ are *different* diagnostics: the residual measures distance to the fixed model direction $\mathbf{1}_N\mathbf{1}_N^t$, while σ_2/σ_1 measures the relative (spectral-norm) distance to the set of all rank-one matrices. A rank-one matrix in a different direction has $\sigma_2/\sigma_1 = 0$ but a large directional residual, so σ_2/σ_1 is only a necessary check; the model at the fixed direction is certified by the directional residual.

A one-sided difference estimate of the jump additionally carries the systematic bias of Proposition 3.3; a centered difference removes it to leading order.

4 What the local event determines: arithmetic rigidity

The first jump recovers $\Lambda(q)$. It is natural to ask whether the higher-order local structure of the event carries any further arithmetic. It does not.

Theorem 4.1 (Arithmetic rigidity of the event). *Let $u_0 = \log q$ be a prime-power edge. For every order $r \ge 1$ the singular jet of the path is*

$$\Delta Q_N^{(r)}(u_0) = \frac{\Lambda(q)}{\sqrt{q}}\, B_{r,N}(u_0),$$

where $B_{r,N}(u_0)$ is a universal matrix depending only on r, N, and the edge location u_0, and not otherwise on q. Consequently, modulo the known edge location, the entire singular event content is the single scalar $\Lambda(q)$: at a fixed edge u_0 the assignment $\Lambda \mapsto (\Lambda/\sqrt{q})\,(B_{r,N}(u_0))_{r\geq 1}$ is a linear injection, so the singular jet family determines and is determined by the single scalar $\Lambda(q)$, and no finer arithmetic invariant of q is present in the local event.

Proof. By Theorem 3.1 and the analyticity of the background, all singular jets at u_0 come from the entering atom $E_q(\tau) = -\frac{\Lambda(q)}{\sqrt{q}} A_N\big(\tau/(u_0+\tau)\big)$, $\tau = u - u_0$: for each r,

$$\Delta Q_N^{(r)}(u_0) = -\frac{\Lambda(q)}{\sqrt{q}}\,\frac{d^r}{d\tau^r}\Big|_{\tau=0} A_N\Big(\frac{\tau}{u_0+\tau}\Big) = \frac{\Lambda(q)}{\sqrt{q}}\,B_{r,N}(u_0),$$

and the bracket depends only on r, N, and u_0. Since $q = e^{u_0}$ is fixed by the edge, $\sqrt{q}$ and every $B_{r,N}(u_0)$ are known; the jets are therefore $\Lambda(q)$ times a known matrix, and $\Lambda(q)$ is read off from any one nonzero entry of $B_{1,N}(u_0) = -2(\log q)^{-1}\mathbf{1}_N\mathbf{1}_N^t$. □

The argument is elementary, but the conclusion is structurally clean: it delimits exactly what the local event can encode. The edge location $u_0 = \log q$ already fixes q, and beyond it the singular jet adds only the scalar $\Lambda(q)$; the event is in this sense a rank-one arithmetic object. The remaining sections show that the geometry *around* the weight, though carrying no further arithmetic, is a sharp and self-contained piece of finite harmonic analysis.

The universal matrices $B_{r,N}$ are not merely well defined; they have an explicit closed form, which makes the rigidity statement concrete and reduces the first two derivative laws to the cases $r = 1, 2$.

Proposition 4.2 (Closed form of the universal jet)**.** *Write $C_j = [\omega^j]A_N(\omega)$ for the jth matrix Taylor coefficient of the single-source matrix; $C_j = 0$ for even j, and for odd $j = 2\ell+1$*

$$(C_j)_{mn} = \frac{(-1)^\ell(2\pi)^j}{j!}\,\frac{m^j - n^j}{\pi(m-n)} \quad (m \neq n), \qquad (C_j)_{mm} = \frac{2(-1)^\ell(2\pi m)^{j-1}}{(j-1)!}.$$

Then the universal matrices of Theorem 4.1 are

$$B_{r,N}(u_0) = -\frac{r!}{u_0^r}\sum_{\substack{1\leq j\leq r\\ j\ \text{odd}}} (-1)^{r-j}\binom{r-1}{j-1} C_j.$$

In particular $B_{1,N} = -2u_0^{-1}\mathbf{1}_N\mathbf{1}_N^t$ (Theorem 3.1) and $B_{2,N} = +4u_0^{-2}\mathbf{1}_N\mathbf{1}_N^t$ (Proposition 3.3).

Proof. By Lemma 2.4, $A_N(\omega) = \sum_{j\ \text{odd}} C_j\,\omega^j$ with the displayed coefficients, read off from the Taylor expansions of $\sin(2\pi m\omega)/\pi$ and $2\omega\cos(2\pi m\omega)$. Compose with $\omega(\tau) = \tau/(u_0+\tau)$ and use the edge transport $[\tau^r]\omega^j = (-1)^{r-j}\binom{r-1}{j-1}u_0^{-r}$ (Lemma 5.3 below); since $B_{r,N} = -r!\,[\tau^r]A_N(\omega(\tau))$ by Theorem 4.1, the sum follows. Evaluating at $r = 1$ ($C_1 = 2\mathbf{1}_N\mathbf{1}_N^t$) and $r = 2$ recovers the first- and second-derivative jumps. □

5 The finite source-to-jet dictionary

We record the finite algebra of the event geometry: how a general finite source measure is encoded in the matrix, and how the edge jets invert that encoding. Sections 5 and 6 are a reference catalog of this finite harmonic analysis, recorded for the event geometry itself. The statements are exact facts about the fixed matrix A_N; by Theorem 4.1 they specialize at a prime event to the recovery of a fixed profile, not to further arithmetic.

Let $\mathcal{V}_N = \operatorname{span}\{b_0, b_k^s, b_k^c : 1 \le k \le N\}$ be the real source-profile space, with

$$b_0(\omega) = \omega, \quad b_k^s(\omega) = \frac{\sin(2\pi k\omega)}{2\pi k}, \quad b_k^c(\omega) = \omega\cos(2\pi k\omega).$$

For $f = ab_0 + \sum_k (A_k b_k^s + B_k b_k^c)$ write $\mathfrak{c}_N(f) = (a, A_1, \dots, A_N, B_1, \dots, B_N)$.

Lemma 5.1 (Finite source-matrix quotient)**.** *For a finite signed measure μ on $[0,1]$ put $B_\mu = \int A_N \, d\mu$ and*

$$M_0 = \int \omega \, d\mu, \quad S_k = \int \sin(2\pi k\omega)\, d\mu, \quad C_k = \int \omega\cos(2\pi k\omega)\, d\mu.$$

Then $\mu \mapsto B_\mu$ factors exactly through the $2N+1$ coordinates (M_0, S_k, C_k), recovered from the entries $(B_\mu)_{00} = 2M_0$, $(B_\mu)_{k0} = S_k/(\pi k)$, $(B_\mu)_{kk} = 2C_{|k|}$; the quotient has dimension exactly $2N+1$. This is the source-to-matrix bijection of [10, Prop. 4.2], specialized to the single-source family used here.

Proof. With the convention $\operatorname{sgn}(0) = 0$, for $m \ne n$ $(B_\mu)_{mn} = (\operatorname{sgn}(m)S_{|m|} - \operatorname{sgn}(n)S_{|n|})/(\pi(m-n))$, and $(B_\mu)_{00} = 2M_0$, $(B_\mu)_{kk} = 2C_{|k|}$; every entry is linear in the listed coordinates. The profiles ω, $\sin(2\pi k\omega)$, $\omega\cos(2\pi k\omega)$ $(1 \le k \le N)$ are linearly independent (exponential polynomials), and the displayed entries invert the coordinates, so the quotient is exactly $2N+1$ dimensional. □

By Lemma 2.4 the profiles carry only odd jets; define the scaled odd edge jets $J_\ell(f) = \frac{(-1)^\ell}{(2\pi)^{2\ell}} f^{(2\ell+1)}(0)$. A direct computation gives $J_\ell(f) = a\delta_{\ell,0} + \sum_{k=1}^N (A_k + B_k(2\ell+1))(k^2)^\ell$.

Theorem 5.2 (Event-jet determinant)**.** *Let M_N be the $d \times d$ matrix, $d = 2N+1$, with rows $\ell = 0, \dots, 2N$ and columns $e_0, X_1, \dots, X_N, Y_1, \dots, Y_N$ given by $M_N(\ell, e_0) = \delta_{\ell,0}$, $M_N(\ell, X_k) = (k^2)^\ell$, $M_N(\ell, Y_k) = (2\ell+1)(k^2)^\ell$. Then*

$$\det M_N = (-1)^{N(N-1)/2}\, 2^N \prod_{k=1}^N k^6 \prod_{1 \le i < j \le N} (j^2 - i^2)^4.$$

In particular $\dim \mathcal{V}_N = 2N+1$ and the first $2N+1$ scaled odd edge jets $J_0, \dots, J_{2N}$ are coordinates on $\mathcal{V}_N$.

Proof. Expand along e_0; in the remaining rows set $r = \ell - 1$. For node $x_k = k^2$ the two columns are x_k^{r+1} and $(2r+3)x_k^{r+1}$; replacing the second by itself minus thrice the first gives $2rx_k^{r+1}$, and factoring x_k and $2x_k^2$ extracts $2^N \prod_k x_k^3 = 2^N \prod_k k^6$. The remaining paired-column matrix is the confluent Vandermonde with each node doubled [13], of determinant $\prod_{i<j}(x_j - x_i)^4$. Reordering paired columns to grouped order costs $N(N-1)/2$ swaps; substituting $x_k = k^2$ gives the formula. □

Lemma 5.3 (Odd edge transport)**.** *For $u_0 > 0$, $\tau = u - u_0$, $\omega = \tau/(u_0 + \tau)$, and integers $r, j \ge 1$, $[\tau^r]\omega^j = (-1)^{r-j}\binom{r-1}{j-1} u_0^{-r}$ for $r \ge j$ and 0 otherwise. On the odd Taylor-jet subspace (orders $1, 3, \dots, 4N+1$) the map from ω-coefficients to τ-coefficients is lower triangular with nonzero diagonal and determinant $u_0^{-(2N+1)^2}$, and likewise for the scaled jets $J_0, \dots, J_{2N}$.*

Proof. $\omega^j = \tau^j u_0^{-j}(1 + \tau/u_0)^{-j}$, and the binomial expansion gives the coefficient; the diagonal $(r = j)$ is u_0^{-j}, and $\sum_{j \text{ odd} \le 4N+1} j = (2N+1)^2$. □

Proposition 5.4 (Observed event stream)**.** *At a detected edge $u_0 = \log q$, the odd matrix jumps $\Delta Q_N^{(1)}, \Delta Q_N^{(3)}, \dots, \Delta Q_N^{(4N+1)}$ determine, for every entry (m,n), the scaled stream $J_0(f_{mn}), \dots, J_{2N}(f_{mn})$ with $f_{mn} = (A_N)_{mn}$; the entries $(0,0)$, $(k,0)$, (k,k) carry the normalized basis $2b_0, 2b_k^s, 2b_k^c$ of the finite quotient.*

Proof. By Theorem 4.1 each odd jump is $(\Lambda(q)/\sqrt{q})$ times the corresponding τ-derivative of $f_{mn}(\tau/(u_0+\tau))$; after dividing by the scalar (known once $\Lambda(q)$ is read from the first jump) and inverting the triangular transport of Lemma 5.3, one recovers the odd ω-coefficients of f_{mn}, and hence $J_\ell(f_{mn})$ by the fixed diagonal scaling. The entry profiles of $(B_\mu)_{00}$, $(B_\mu)_{k0}$, $(B_\mu)_{kk}$ are $2b_0, 2b_k^s, 2b_k^c$ by Lemma 5.1. □

Theorem 5.5 (Finite dictionary: window, blind line, recurrence)**.** *Let $m_\ell = a\delta_{\ell,0} + \sum_{k=1}^{N}(A_k + B_k(2\ell+1))(k^2)^\ell$. Then:*

(i) the first $2N+1$ values $m_0,\dots,m_{2N}$ recover $(a, A_1,\dots,A_N, B_1,\dots,B_N)$ by exact inversion of M_N;

(ii) the first $2N$ values leave a one-dimensional blind line, on which m_{2N} is nonzero (the window $2N+1$ is sharp);

(iii) the whole stream is annihilated by $P_N(S) = S\prod_{k=1}^{N}(S-k^2)^2$, $(Sm)_\ell = m_{\ell+1}$, and this order is sharp on the full quotient.

Proof. (i) is Theorem 5.2. For (ii), the d rows of M_N are independent, so the first $d-1$ have a one-dimensional nullspace; a vector in it with $m_{2N}=0$ would lie in $\ker M_N$, impossible. For (iii), S annihilates $\delta_{\ell,0}$ and, for $x=k^2$, $(S-x)x^\ell = 0$ and $(S-x)^2((2\ell+1)x^\ell) = 0$; a shorter recurrence would make the first $2N+1$ rows of M_N dependent, contradicting Theorem 5.2. □

Thus the source-to-jet map is a lossless finite dictionary with prescribed nodes $0, 1^2, \dots, N^2$ and sharp window $2N+1$ (Figure 2). This is the finite Prony structure [12, 13, 6] of the specific matrix A_N; by rigidity it recovers, at a prime event, the fixed profile and the scalar $\Lambda(q)$, not further arithmetic.

6 A finite vanishing-moment ceiling at the prime edge

How blind can a band-limited probe be to an entering prime? For a real even vector $x = (x_m)_{m\in I_N}$, let $T_x(t) = \sum_m x_m e^{2\pi i m t}$, let $M_j(x) = \sum_m m^j x_m$ be its moments (odd moments vanish by evenness), and let $e(x) = \min\{j \ge 0 : M_j(x) \ne 0\}$ be its first nonvanishing (even) moment. The Rayleigh response of x to the entering atom, $x^t A_N(\omega)x = K_x(\omega)$ with $K_x(\omega) = 2\int_0^\omega T_x(t)T_x(\omega-t)\,dt$, has vanishing order $2e(x)+1$ in ω, hence, via Lemma 5.3, in the cutoff variable τ; we call $2e(x)+1$ the *visibility order* of x. The bound below is a vanishing-moment (sum-rule) ceiling; the phrase *prime-edge uncertainty principle* is kept only as a gloss for its band-limited reading.

Theorem 6.1 (Finite vanishing-moment ceiling at the prime edge)**.** *For every nonzero real even x on I_N the visibility order satisfies $2e(x)+1 \le 4N+1$. Equality holds if and only if, up to a nonzero scalar, x is the centered finite-difference stencil*

$$x_m = (-1)^{N-m}\binom{2N}{N+m}, \qquad -N \le m \le N,$$

which is therefore the unique maximally blind band-limited normal, blind to order $4N+1$. Moreover $K_x(\omega) = \dfrac{2(2\pi)^{2e}M_e(x)^2}{(2e+1)!}\,\omega^{2e+1} + O(\omega^{2e+3})$, $e = e(x)$.

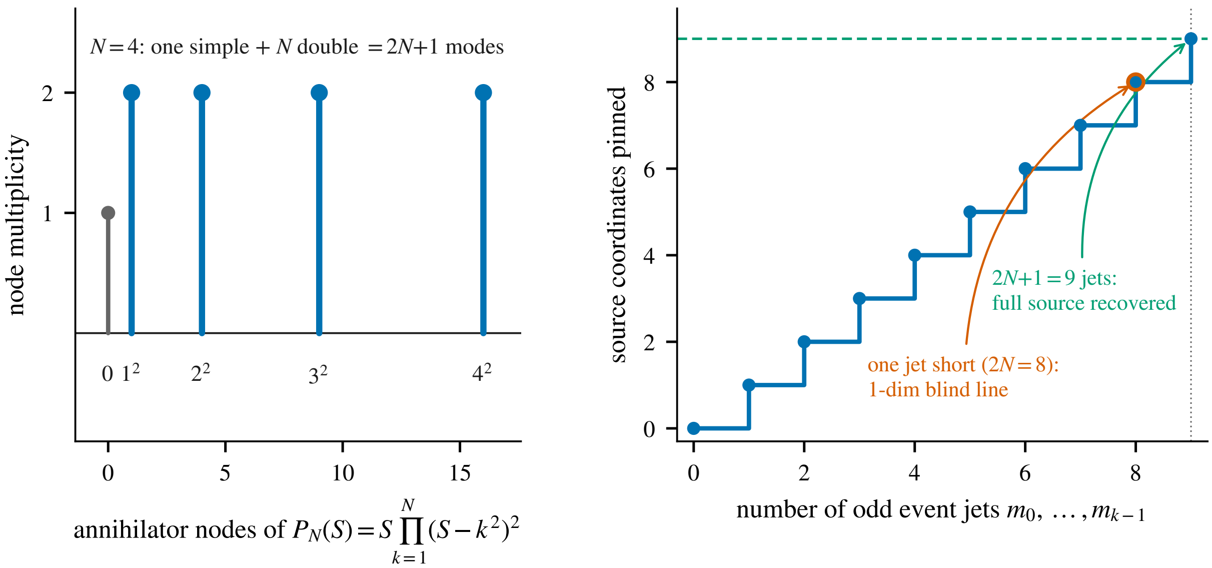

Figure 2: The finite source-to-jet dictionary at level $N = 4$. Left: the annihilator nodes of $P_N(S) = S\prod_{k=1}^{N}(S-k^2)^2$, one simple node at 0 and N double nodes at $1^2, \dots, N^2$, totalling $2N{+}1$ modes. Right: the number of source coordinates pinned down by the first k odd jets saturates at $2N{+}1 = 9$; one jet short leaves the sharp one-dimensional blind line.

Proof. The Volterra expansion gives $[\omega^{k+1}]K_x(\omega) = \frac{2(2\pi i)^k}{(k+1)!}\sum_{a=0}^{k} M_a(x)M_{k-a}(x)$; with odd moments zero and $e = e(x)$ the first nonzero even moment, the leading term is at $k = 2e$, giving order $2e{+}1$ and the stated coefficient: since all odd moments vanish, e is even, so $i^{2e} = (-1)^e = 1$ and the coefficient is real and positive. The even moments $M_0, M_2, \dots, M_{2N}$ of an even vector are the image of $(x_0, \dots, x_N)$ under a Vandermonde matrix in the squared nodes $\{0^2, 1^2, \dots, N^2\} = \{0, 1, 4, \dots, N^2\}$, which is invertible; hence they cannot all vanish for $x \neq 0$, so $e(x) \leq 2N$. Requiring $M_0 = \cdots = M_{2N-2} = 0$ imposes N independent conditions on the $N+1$ even degrees of freedom, leaving a one-dimensional solution; the centered stencil satisfies them with $M_{2N} = (2N)! \neq 0$, and is therefore the unique extremizer up to scale. This is the classical vanishing-moment (sum-rule) bound [11], here realized as the prime-edge visibility ceiling of the finite Connes–van Suijlekom path. □

Figure 3 shows the extremal stencil and the visibility ceiling. The ceiling is sharp and the extremizer unique: a level-N band-limited probe cannot be blind to an entering prime beyond order $4N + 1$, and exactly one probe (the $2N$-th central difference) attains that bound.

The integer ceiling has a metric refinement. On a band-limited window the leading term of Theorem 6.1 controls the response, and converts the visibility *order* into a visibility *scale*.

Corollary 6.2 (Edge resolution scale)**.** *Fix a detected edge $u_0 = \log q$ and a nonzero real even x with first nonvanishing even moment $e = e(x)$. Then there is $\omega_*(x) > 0$ on which K_x is strictly positive and strictly increasing, and, transporting to the cutoff variable $\tau = u - u_0$ by $\omega = \tau/(u_0 + \tau)$, the advance past the edge at which $|K_x|$ first reaches a threshold ε satisfies, as $\varepsilon \to 0$,*

$$\tau_\varepsilon(x) = \log q\left(\frac{(2e{+}1)!\,\varepsilon}{2(2\pi)^{2e}M_e(x)^2}\right)^{1/(2e+1)}(1 + o(1)).$$

The first-passage exponent $1/(2e(x) + 1)$ decreases with the visibility order, so for small ε the

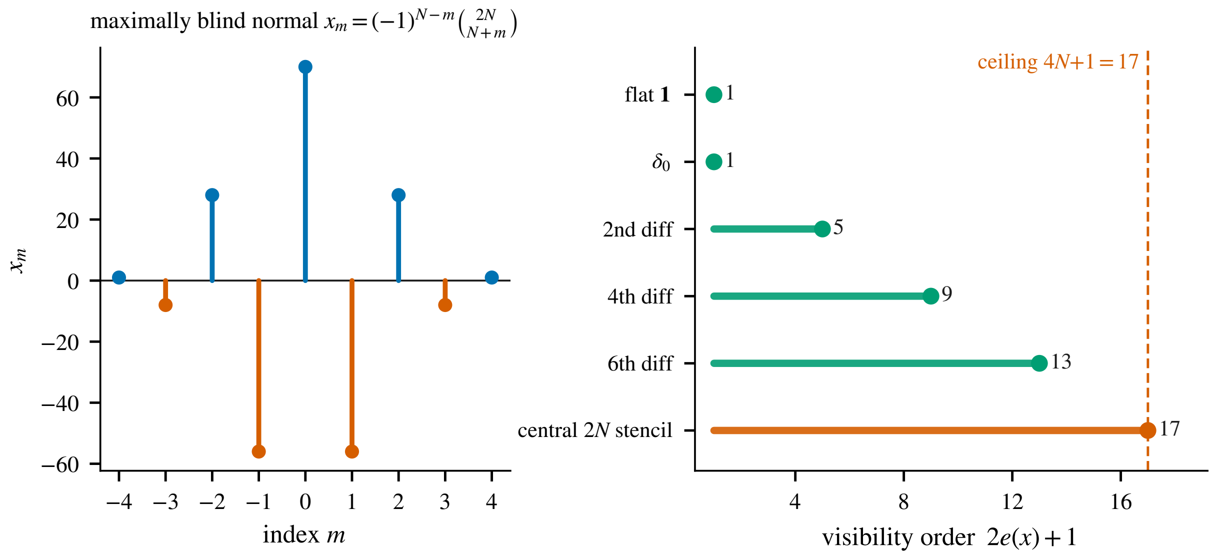

Figure 3: The finite vanishing-moment ceiling at the prime edge, shown at $N = 4$. Left: the unique maximally blind normal, the centered finite-difference stencil $x_m = (-1)^{N-m}\binom{2N}{N+m}$. Right: the visibility order $2e(x) + 1$ of a family of even normals climbs by 4 with each additional vanishing moment and is capped at $4N + 1 = 17$, attained only by the central-difference stencil.

maximally blind probe $e = 2N$ has the coarsest edge resolution and the generic probe $e = 0$ the finest: the integer ceiling of Theorem 6.1 is the order of this resolution scale.

Proof. By Theorem 6.1, $K_x(\omega) = \frac{2(2\pi)^{2e}M_e^2}{(2e+1)!}\omega^{2e+1}(1+O(\omega^2))$ with a positive leading coefficient. The remainder is an analytic multiple of ω^{2e+3}, so it is dominated by the leading term on a nonempty window $0 < \omega \le \omega_*(x)$ (such a window exists because the remainder is $O(\omega^{2e+3})$ with the leading term of exact order ω^{2e+1}; the window scale is set by the ratio of successive even moments of x); there the leading term controls sign, monotonicity, and, as $\varepsilon \to 0$, the first-passage scale. Inverting $K_x(\tau) = \varepsilon$ with $\omega = \tau/\log q + O(\tau^2)$ gives τ_ε. The window $\omega_*(x)$ depends on x through those moment ratios and is not uniform over the unit sphere. $\square$

Remark 6.3. The integer $2e(x) + 1$ is upper semicontinuous and generically equal to 1: every neighborhood of the extremal stencil contains probes of visibility order 1, so the ceiling is attained only on a measure-zero set. The resolution scale τ_ε is the robust, metric form of the same statement.

7 The coincidence-resolvent generating identity

The first-derivative jump is a rank-one negative shock in the fixed universal direction $\mathbf{1}_N$. Contracting the resolvent against that direction generates, in one rational function, the jump of every spectral flow: the log-determinant, every trace moment, the eigenvalue velocities, and the elementary barriers.

Theorem 7.1 (Coincidence-resolvent generating identity)**.** *Let $u_0 = \log q$ be a prime-power event, $Q_0 = Q_N(u_0)$, and $a_q = 2\Lambda(q)/(\sqrt{q}\,\log q) > 0$. Define the* coincidence resolvent

$$G_N(z) = \langle \mathbf{1}_N, (Q_0 - z)^{-1}\mathbf{1}_N \rangle = \sum_{j=1}^{d} \frac{|\langle \phi_j, \mathbf{1}_N\rangle|^2}{\lambda_j - z}, \qquad z \notin \mathrm{spec}(Q_0),$$

a rational function whose poles lie among the eigenvalues of Q_0: grouping the sum by spectral projections P_λ, $G_N(z) = \sum_{\lambda\in\mathrm{spec}(Q_0)} \|P_\lambda \mathbf{1}_N\|^2/(\lambda - z)$ has a simple pole with residue $-\|P_\lambda \mathbf{1}_N\|^2$ at exactly those eigenvalues λ with $P_\lambda \mathbf{1}_N \neq 0$; an eigenvalue whose eigenspace is orthogonal to $\mathbf{1}_N$ contributes no pole (the corresponding point is removable), and a repeated eigenvalue contributes one grouped residue, not one pole per basis eigenvector. Then, at u_0, for $z \notin \mathrm{spec}(Q_0)$ (only the derivative jump enters, not the branch of $\log\det$),

$$\Delta \frac{d}{du} \log\det(Q_N(u) - z) = -a_q\, G_N(z), \qquad \Delta \frac{d}{du} \operatorname{tr} Q_N(u)^p = -p\, a_q \left\langle \mathbf{1}_N, Q_0^{p-1}\mathbf{1}_N\right\rangle \ \ (p \geq 1).$$

In particular $\Delta \frac{d}{du} \operatorname{tr} Q_N(u_0) = -a_q\, d$, so the scalar trace path $\operatorname{tr} Q_N$ has singular curvature $(D_u^2 \operatorname{tr} Q_N)_{\mathrm{sing}} = -d \sum_q a_q \delta_{\log q}$: its singular second derivative is exactly $-(2N+1)$ times the scalar von Mangoldt measure μ_{mn} of Corollary 3.2, hence carries the prime side of the explicit formula under the same conversion.

Proof. Q_N is continuous at u_0, so $(Q_N(u) - z)^{-1}$ and $Q_N(u)^{p-1}$ are two-sided continuous there; the only discontinuity is the velocity jump $\Delta Q_N'(u_0) = -a_q \mathbf{1}_N \mathbf{1}_N^t$ (Theorem 3.1). By Jacobi's formula $\frac{d}{du} \log\det(Q_N - z) = \operatorname{tr}((Q_N - z)^{-1} Q_N')$, whose jump is $\operatorname{tr}((Q_0 - z)^{-1} \Delta Q_N') = -a_q \mathbf{1}_N^t (Q_0 - z)^{-1}\mathbf{1}_N$; the spectral decomposition gives G_N. Likewise $\frac{d}{du} \operatorname{tr} Q_N^p = p \operatorname{tr}(Q_N^{p-1} Q_N')$ has jump $-p\, a_q\, \mathbf{1}_N^t Q_0^{p-1} \mathbf{1}_N$; for $p = 1$ this is $-a_q \|\mathbf{1}_N\|^2 = -a_q d$. □

The residues of G_N are $-|\langle \phi_j, \mathbf{1}_N\rangle|^2$; their magnitudes are the couplings to the universal direction, and these magnitudes, scaled by $-a_q$, are exactly the eigenvalue-velocity jumps of Corollary 7.3.

Remark 7.2 (The coincidence spectral measure). Writing $\rho = \sum_j |\langle \phi_j, \mathbf{1}_N\rangle|^2\, \delta_{\lambda_j}$ for the finite positive measure carried by the couplings, $G_N(z) = \int (\lambda - z)^{-1}\, d\rho$ is its Cauchy transform and every trace-moment jump is a moment of ρ: $\Delta \frac{d}{du} \operatorname{tr} Q_N^p = -p\, a_q \int \lambda^{p-1}\, d\rho$, with $\rho(\mathbb{R}) = \|\mathbf{1}_N\|^2 = 2N + 1$ and $\int \lambda\, d\rho = \langle \mathbf{1}_N, Q_0 \mathbf{1}_N\rangle$. Thus a single finite measure governs all the spectral-flow jumps at the event.

Corollary 7.3 (Velocity deflation and interlacing)**.** *The velocity spectrum deflates by rank one: $Q_N'(u_0{+}0) = Q_N'(u_0{-}0) - a_q \mathbf{1}_N \mathbf{1}_N^t$, so (Weyl/Cauchy) the right velocity spectrum interlaces the left,*

$$\lambda_1\big(Q_N'(u_0{+}0)\big) \leq \lambda_1\big(Q_N'(u_0{-}0)\big) \leq \lambda_2\big(Q_N'(u_0{+}0)\big) \leq \cdots \leq \lambda_d\big(Q_N'(u_0{-}0)\big),$$

and, for simple Q_0, each eigenvalue velocity decelerates by $\Delta\lambda_j'(u_0) = -a_q |\langle \phi_j, \mathbf{1}_N\rangle|^2 \leq 0$ (the residues of G_N), with total $\sum_j \Delta\lambda_j' = -a_q d$, strict off $\mathbf{1}_N^\perp$.

Proof. Immediate from $\Delta Q_N'(u_0) = -a_q \mathbf{1}_N \mathbf{1}_N^t$ and, for the velocities, first-order perturbation $\Delta\lambda_j' = \langle \phi_j, \Delta Q_N' \phi_j\rangle$. □

Remark 7.4 (Elementary spectral barriers in a positive cell; conditional). Suppose $Q_0 = Q_N(u_0)$ is positive definite. (This is a hypothesis, not a verified property of the actual Connes–van Suijlekom path at the program's canonical parameters, where the assembled form has near-zero eigenvalues [16]; whether any actual prime-power event sits in a positive cell is not established here, and the statement is conditional on it, which is why it is recorded as a remark rather than a theorem.) Let $e_r(Q)$ be the rth elementary symmetric function of the eigenvalues of Q. Then for every $1 \leq r \leq d$,

$$\Delta \frac{d}{du} e_r(Q_N(u))\Big|_{u=u_0} = -a_q\, \mathbf{1}_N^t T_{r-1}(Q_0) \mathbf{1}_N < 0, \qquad T_{r-1}(Q_0) = \sum_{j=0}^{r-1} (-1)^j e_{r-1-j}(Q_0) Q_0^j.$$

In particular the determinant barrier ($r = d$) and its logarithm strictly decelerate at every prime event in such a cell.

Proof of the barrier inequalities. For a principal r-set S, the matrix determinant lemma gives $\det(Q_{0,S} - ta_q \mathbf{1}_S \mathbf{1}_S^t) = \det Q_{0,S} - ta_q\, \mathbf{1}_S^t \operatorname{adj}(Q_{0,S})\mathbf{1}_S$; $Q_{0,S}$ is positive definite so $\operatorname{adj}(Q_{0,S})$ is positive definite and each summand is positive; summing over $|S| = r$ proves strict negativity. The Newton-transformation identity $De_r(Q)[H] = \operatorname{tr}(T_{r-1}(Q)H)$ with $H = -a_q \mathbf{1}_N \mathbf{1}_N^t$ gives the displayed value; in an eigenbasis of Q_0, $T_{r-1}(Q_0)$ is diagonal with positive entries, so the quadratic form on $\mathbf{1}_N$ is positive. □

The prime event as a rank-one Weyl-function increment

The rank-one velocity deflation has a clean resolvent reading. A related scalar-resolvent reduction appears in Andrade's analysis of the pole term of the continuum localized form [4], where a rank-one-per-parity splitting reduces a bottom-eigenvalue question to a scalar resolvent criterion with a Krein analysis; the object here is different, namely the prime-event velocity jump of the fixed-N path rather than the pole term. Let $V_\pm = Q_N'(u_0 \pm 0)$ be the right and left spectral velocities at $u_0 = \log q$, so $V_+ = V_- - a_q \mathbf{1}_N \mathbf{1}_N^t$ by Theorem 3.1, and define the *coincidence-Weyl function* of the velocity in the universal direction,

$$W_\pm(z) = \left\langle \mathbf{1}_N, (V_\pm - z)^{-1} \mathbf{1}_N \right\rangle, \qquad z \notin \operatorname{spec}(V_\pm).$$

$W_\pm$ is a rational Nevanlinna function; when $\mathbf{1}_N$ is cyclic for $V_\pm$ it is $(2N+1)$ times the Weyl function of the finite Jacobi matrix obtained by tridiagonalizing $V_\pm$ from the normalized vector $\mathbf{1}_N/\sqrt{2N+1}$ (the identities below need no such assumption).

Theorem 7.5 (Prime event as a rank-one Weyl increment)**.** *For every z off the spectra of $V_\pm$, $1 - a_q W_-(z) \neq 0$ and*

$$W_+(z) = \frac{W_-(z)}{1 - a_q W_-(z)}, \qquad a_q = \frac{2\Lambda(q)}{\sqrt{q}\,\log q} > 0.$$

Equivalently, as an identity of meromorphic functions,

$$\frac{1}{W_+(z)} = \frac{1}{W_-(z)} - a_q,$$

valid pointwise exactly where $W_+(z)W_-(z) \neq 0$; the constant decrement is independent of z. *(The pointwise restriction is not vacuous: a nonconstant real rational Nevanlinna function has real zeros interlacing its poles, so $W_\pm$ generically vanish at points off both spectra whenever $\mathbf{1}_N$ has nonzero projection onto at least two eigenspaces, the exceptional case being an interlacing zero that coincides with an eigenvalue whose eigenspace is orthogonal to $\mathbf{1}_N$, and at such points off the spectra both reciprocals are undefined.) Equivalently V_+ is the Schur complement, with respect to the scalar block, of the bordered matrix $\begin{pmatrix} a_q^{-1} & \mathbf{1}_N^t \\ \mathbf{1}_N & V_- \end{pmatrix}$, i.e. a single symmetric elimination step with pivot $a_q^{-1} = \sqrt{q}\,\log q/(2\Lambda(q))$. Consequently the von Mangoldt weight is read off from the reciprocal increment,*

$$\Lambda(q) = \frac{\sqrt{q}\,\log q}{2}\left(\tfrac{1}{W_-(z)} - \tfrac{1}{W_+(z)}\right) \qquad \textit{wherever } W_+(z)W_-(z) \neq 0.$$

Proof. For z off both spectra, $\det(V_+ - z) = \det(V_- - z)\,(1 - a_q W_-(z))$ by the matrix determinant lemma, so $1 - a_q W_-(z) \neq 0$ there. The Sherman–Morrison identity applied to $V_+ = V_- - a_q \mathbf{1}_N \mathbf{1}_N^t$ with $M = V_- - z$ gives $W_+ = \mathbf{1}_N^t (M - a_q \mathbf{1}_N \mathbf{1}_N^t)^{-1} \mathbf{1}_N = W_-/(1 - a_q W_-)$. Dividing by $W_+ W_-$ where both are nonzero yields the reciprocal form, which then extends to the stated meromorphic identity. The Schur statement is the identity $V_- - \mathbf{1}_N\, a_q\, \mathbf{1}_N^t = V_+$ for the displayed border. The zero/pole interlacing for a real rational Nevanlinna function is classical [19] ($W_-' > 0$ between consecutive real poles). □

Remark 7.6 (Krein-string analogy, not established here). The constant decrement of $1/W$ is *formally* the boundary point-mass move in Krein's string–to–continued-fraction correspondence [18], and it is tempting to read a_q as a per-event boundary mass in the same de Branges–Krein canonical-system setting as Suzuki's continuous screw-function pencil [25], with the reduced parametrization $(\omega_q'(u_0) = 1/\log q)$ matching the prime data his continuous screw function carries. Making that reading literal, however, requires Stieltjes-class hypotheses (a positive string with spectral measure supported on a half-line [18]) that the indefinite velocities $V_\pm$ are not proved to satisfy here; a general rational Nevanlinna function is the Weyl function of a finite Jacobi matrix or canonical system [14, 19], which is a strictly larger class than the Krein strings. We therefore record the string reading as an analogy, not a theorem. In any case only the *per-event* reciprocal increment equals a_q; the interior structure of V_- encodes its accumulated spectral history, not the individual primes, and nothing here bears on positivity of the finite form or on the zeros.

8 Universality, the explicit formula, and scope

Universality across the Selberg class. The transfer below is conditional on the construction of the corresponding finite path: it applies to any finite path assembled with the same prime-block template as the Connes–van Suijlekom path, with the ζ source weights replaced by those of $L(s,\pi)$; such a path is exhibited here only for ζ. For Dirichlet twists the prime block is written down entrywise below, but no full twisted path (in particular no twisted archimedean or pole block) is constructed, and the construction for a general Selberg-class L-function is not carried out in this paper. Granting the path, the prime block is determined only by the source amplitudes and the test-function-at-$\log q$ pairing: the single-source matrix $A_N(1-\log q/u)$ is the same divided-difference matrix regardless of the L-function, since the gamma factor enters only the archimedean block Q_{arch}, not the prime term. For an L-function $L(s,\pi)$ with prime side $\sum_q c_\pi(q)q^{-1/2}\phi(\log q)$ (Dirichlet $c_\chi(q) = \Lambda(q)\chi(q)$, or automorphic $c_\pi(q) = \Lambda(q)a_\pi(q)$), such a finite path has first-derivative jump $-2c_\pi(q)(\log q)^{-1}q^{-1/2}\mathbf{1}_N\mathbf{1}_N^t$, and the whole event *algebra* transfers verbatim: the rank-one direction $\mathbf{1}_N\mathbf{1}_N^t$, the source-to-jet dictionary on the squared nodes $\{0, 1^2, \ldots, N^2\}$, and the vanishing-moment ceiling depend only on A_N and hold for arbitrary (in particular complex) weights. The *sign-definite* consequences are different: negative semidefiniteness of the event measure (Corollary 3.2), the velocity interlacing and deceleration (Corollary 7.3), and the strict barrier decrease (Remark 7.4) all use $a_q > 0$, hence a *positive* weight $c_\pi(q) > 0$. This holds at every prime power for ζ (where $c(q) = \Lambda(q) > 0$), so those results transfer verbatim there, the barrier statement remaining conditional on its positive-definiteness hypothesis (Remark 7.4). For a real primitive character the atom is Hermitian, but $c(q) = \Lambda(q)\chi(q)$ is negative wherever $\chi(q) = -1$ (for instance $\chi_4(3) = -1$), and there the three sign-definite statements reverse; they hold only at prime powers with $\chi(q) > 0$. For complex $c_\pi(q)$ the atom is not even Hermitian. For Dirichlet twists the family of first jumps over χ mod m separates the edge by residue class through character orthogonality, a finite rendering of the prime side of the twisted explicit formula [17].

Corollary 8.1 (Residue-class readout across the Dirichlet family). *Fix a prime-power edge $u_0 = \log q$ with $(q,m) = 1$. Assume, for each Dirichlet character χ mod m, a finite path with the same prime-block template, carrying the twisted weights $c_\chi(q') = \Lambda(q')\chi(q')$, whose non-prime blocks are real analytic at u_0; such paths are granted, not constructed, here. Then the first-derivative jump of the χ-path is $J^{(\chi)} = j^{(\chi)}\mathbf{1}_N\mathbf{1}_N^t$ with $j^{(\chi)} = -2\Lambda(q)\chi(q)(\log q)^{-1}q^{-1/2}$ (by the argument of Theorem 3.1),* *and for every residue a with $(a,m)=1$,*

$$-\frac{\sqrt{q}\,\log q}{2}\cdot\frac{1}{\phi(m)}\sum_{\chi \bmod m}\overline{\chi(a)}\,j^{(\chi)}=\Lambda(q)\,\mathbf{1}[\,q\equiv a \pmod{m}\,].$$

The already-localized weight $\Lambda(q)$ is thus resolved by residue class from the joint first jumps of the Dirichlet family; this reads $\Lambda(q)$ in an arithmetic progression at a known edge, and is not a prediction of q.

Proof. The jump formula follows exactly as in Theorem 3.1: the analytic non-prime blocks cancel in the right-minus-left derivative, and the entering atom carries the twisted weight. Then substitute $j^{(\chi)}$ and apply the character orthogonality $\frac{1}{\phi(m)}\sum_{\chi \bmod m}\overline{\chi(a)}\chi(q)=\mathbf{1}[q\equiv a \pmod{m}]$, valid for $(q,m)=1$. □

The prime side of the explicit formula. Corollary 3.2 places the primes as the singular second derivative of Q_N: the finite path carries the *prime side* of the Weil–Guinand explicit formula [26, 17] exactly and without an archimedean cutoff, as a matrix-valued measure. This is by construction (the prime sum is a defining summand of Q_N), and we prove nothing here about the zero side: the identity relating the prime side to the zeros is the classical explicit formula, supplied externally by the archimedean and pole blocks of Paper 2 [15] and the eigenvalue-to-zero approximation of Paper 1 [16] (with a quantitative convergence law for the first zero in the CCM setting in [5]). The object is thus complementary to Suzuki's continuous screw-function picture [25], to the CCM spectral triple [9], and to Liu's determinant encoding of local Euler factors [21], each a differently mechanized realization of the same arithmetic.

Scope. The claims are finite and structural. The paper proves no positivity of the finite form, no Riemann Hypothesis, no prime-counting or next-prime bound, and no factoring statement; by Theorem 4.1 the local event carries exactly $\Lambda(q)$, and the reconstruction, vanishing-moment, and deflation results are finite harmonic analysis of the fixed path, not arithmetic prediction. The event jets are right-minus-left jumps taken after the common analytic background has cancelled.

Verification. The proofs are algebraic; the companion scripts are separate reproducibility checks for signs, indexing, determinant factors, and finite-field behavior, not proof substitutes. The ancillary verification package records (scripts and JSON artifacts listed in its manifest, each script writing the JSON artifact of the same stem, e.g. `rank_one_jump_audit.json`), over exact integer or modular arithmetic:

rank-one jump, end-to-end	$q\in\{3,4,5,7,8,9,25\}$, $N\le 6$	`check_rank_one_jump`
event-jet and transport rank	$N\le 200$, four prime fields	`check_event_jet_largeN`
event-jet and transport determinants	$N\le 200$, four prime fields	`check_event_jet_determinant`
recurrence	$N\le 1000$, four prime fields	`check_event_jet_recurrence`
event dictionary (window, blind line)	$N\le 200$	`check_event_prony_reconstruction`
source quotient and transport	$N\le 200$	`check_source_quotient_and_transport`
vanishing-moment ceiling (extremizer)	$N\le 10$	`check_uncertainty_ceiling`
spectral-barrier deceleration	50 positive definite cases	`check_spectral_barrier_jump`
universal jet $B_{r,N}$ closed form	exact symbolic, orders $r\le 5$	`check_universal_jet`
coincidence-averaged readout	$(2N+1)^2$ variance (IID model)	`check_coincidence_readout`
Dirichlet residue-class readout	cyclic moduli $m\le 13$	`check_dirichlet_readout`

As a floating-point check on the actual assembled operator at the program's canonical scale, at $N=200$ (dimension 401) every prime power $q\le 100$ was confirmed to have a rank-one first-derivative jump equal to $-2\Lambda(q)/(\sqrt{q}\,\log q)\mathbf{1}_N\mathbf{1}_N^t$: over all 35 events the second singular value of the jump is at most 4×10^{-7} of the first, the jump matches $-a_q\mathbf{1}_N\mathbf{1}_N^t$ to relative error 10^{-6}, and $\Lambda(q)$ is recovered from a single entry to 5×10^{-8}; these four figures are produced by `check_canonical_scale.py` and recorded in its artifact `canonical_scale_audit.json` in the ancillary archive. The archimedean and pole blocks (Lemma 2.2) are analytic and cancel in the jump, so the prime part carries the entire singular event.

Statement on use of AI tools. Language-model tools were used as a drafting and verification-tooling assistant under the author's direction; the author takes sole responsibility for all mathematics, computations, and text.

Data and code availability. The verification package (the separate reproducibility check scripts, their exact JSON artifacts, and the figure generator) accompanies this paper as ancillary files; a checksummed archive is deposited on Zenodo, concept DOI 10.5281/zenodo.21242028 (which resolves to the latest version).

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